

Purely right-censored data with finite observation window

- ▶ Suppose $(X_i, \Delta_i, Z_i), i = 1, \dots, n$, are observed in the study course period $[0, \tau]$. If $X_i > \tau$ then replace X_i by τ ; that is, we study the behavior of $T^* = \min(T, \tau)$.
- ▶ Nonparametric estimation of the joint distribution $\hat{F}(t, z) = \hat{P}(T \leq t, Z \leq z)$.
- ▶ Use a model $P(T|Z)$ (parametric model, Cox, etc.) to get the survival function $S(t; Z)$. Define $U = S(T; Z)$.
- ▶ Use data generated by $\hat{S}(t, z), (T_1, Z_1), \dots, (T_n, Z_n)$, to calculate $U_1, \dots, U_n$. With $S(t; Z) = P(T > t; Z)$, note that $S(\tau; Z)$ represents the long-term survival probability.
- ▶ Find upper bound and lower bound L and R from $\{U_1, \dots, U_n\}$. Note that these two bounds are quite 'model-free.'
- ▶ Under the Cox PHM, $\hat{S}(t; Z) = \exp\{-\hat{\Lambda}(t)\exp(Z_{n+1}\hat{\beta})\}$.
- ▶ Construct the conformal interval by solving the following inequality for T_{n+1} using Z_{n+1} :

$$L \leq \exp\{-\hat{\Lambda}(T_{n+1})\exp(Z_{n+1}\hat{\beta})\} \leq R$$
$$\hat{\Lambda}^{-1}\{-(\ln R)e^{-Z_{n+1}\hat{\beta}}\} \leq T_{n+1} \leq \hat{\Lambda}^{-1}\{-(\ln L)e^{-Z_{n+1}\hat{\beta}}\}$$

- ▶ Remark 1: Depending on the proportion of long-term survivors, there might be a good proportion of $\hat{\Lambda}^{-1}\{-(\ln L)e^{-Z_{n+1}\hat{\beta}}\}$ (left-end point from last interval) equal to or exceeding τ .